Lab 6: Code and Branch Coverage

Coffee Shop Order | HEMA SUSHMITHA N | PES1UG24CS667

Objective and scope

Measure the original test coverage and improve it to at least 90% by adding test cases. The original **coffee_shop_order.py** application and starter test were preserved unchanged. Both original files were verified byte for byte against the assigned source.

Work completed

1. Checked Python, Git and VS Code; created an isolated Python 3.14.6 environment with pytest 9.2.0 and coverage 7.17.0.
2. Ran the starter test for initial line and branch coverage.
3. Added separate line and branch test suites, asserted expected totals, statuses and exceptions, and reran coverage against the application only.
4. Saved final terminal screenshots, text and JSON reports; submitted two coverage commits in the existing SE repository under **Lab-6**.

Measured results

Suite	Passing tests	Statement coverage	Branch outcomes
Original starter	1	26/54 (48.15%)	14/38 (36.84%)
Code coverage tests	8	54/54 (100%)	Not measured in line-only run
Branch coverage tests	39	54/54 (100%)	38/38 (100%)

All **48 tests pass** together (1 starter + 8 line + 39 branch). In a branch-enabled report, the Cover column combines statement and branch outcomes: the initial displayed 43% is (26 + 14) / (54 + 38), rather than the branch-only percentage.

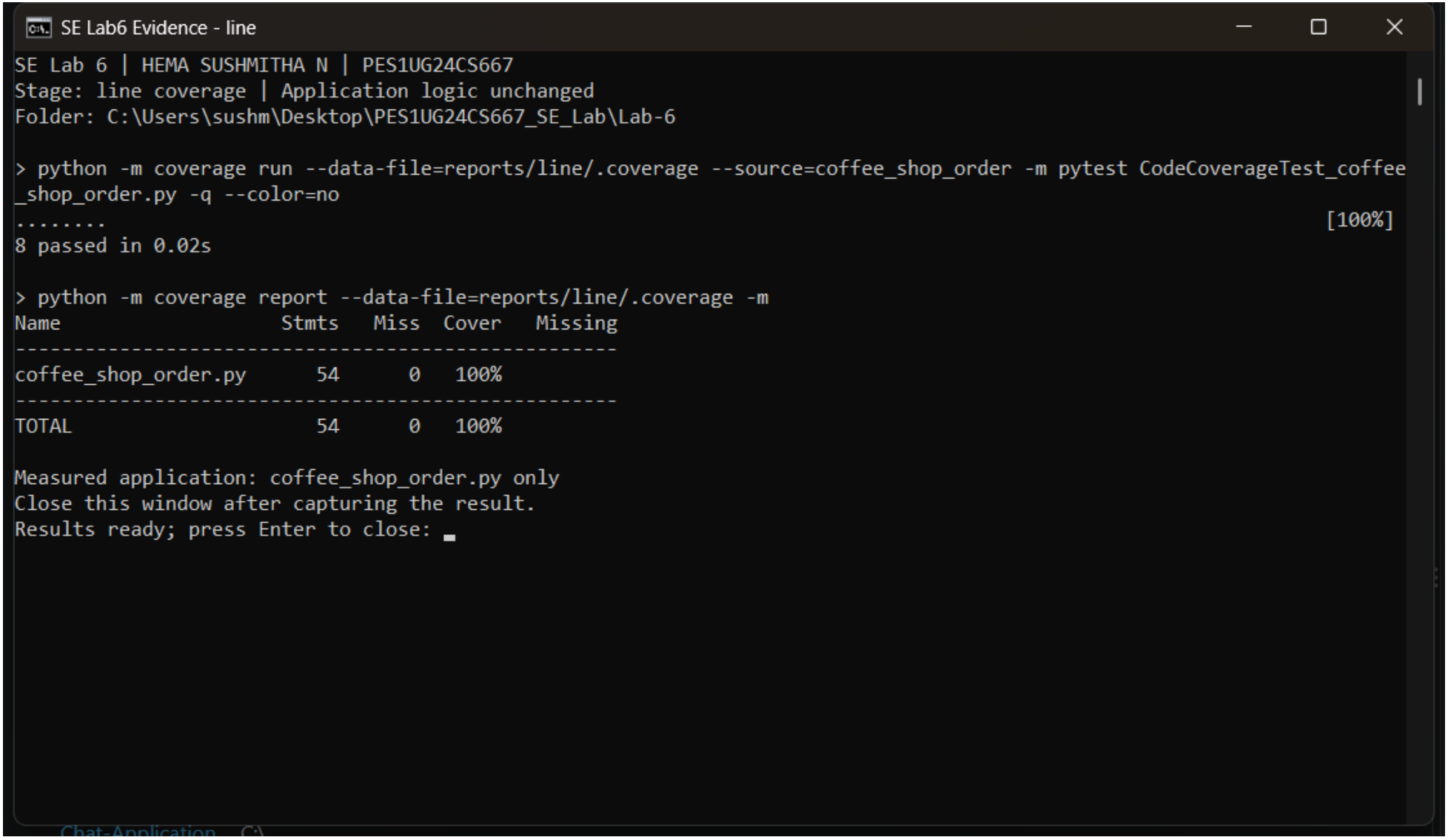
Test coverage and submission

Cases include invalid input, paid/unpaid orders, sizes, defaults, membership, tax, syrup cap boundaries, negative costs, strict/non-strict allergies, four/five rejection thresholds, and machine temperature boundaries and streak resets. Final results meet the handout target; 100% coverage does not establish correctness for all possible inputs.

Assigned source | Submission folder | Line commit: d86bbb7 | Branch commit: 121ac8d

Deliverable 1: Code Coverage Evidence

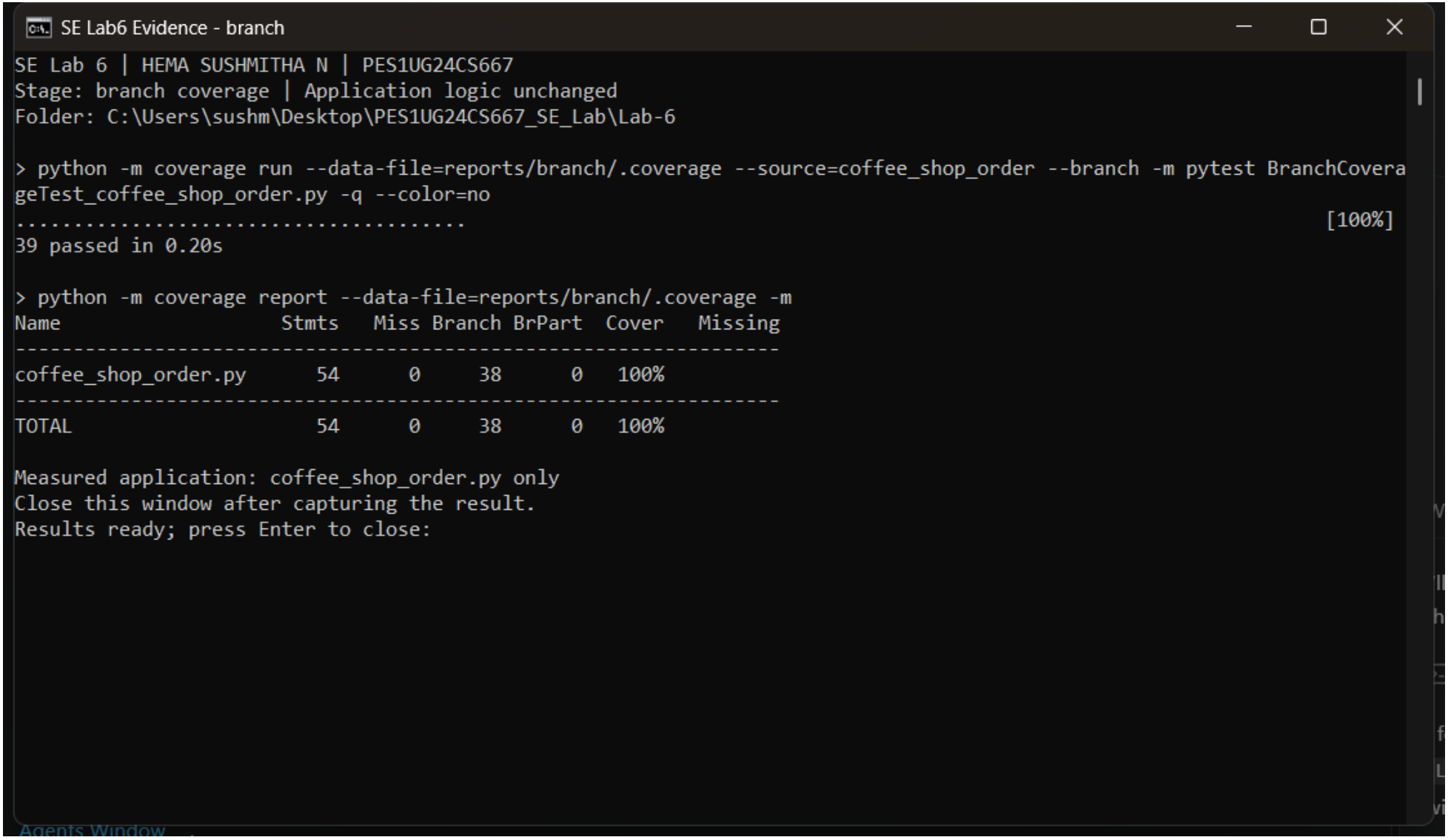
CodeCoverageTest_coffee_shop_order.py | 8 passed | 54/54 statements | 100%



Actual terminal screenshot from the submission checkout. Coverage is limited to coffee_shop_order.py; no application statements are missed.

Deliverable 2: Branch Coverage Evidence

BranchCoverageTest_coffee_shop_order.py | 39 passed | 38/38 branch outcomes | 100%



Actual terminal screenshot from the submission checkout. Both statement and branch coverage are 100%, with no missing statements or partially covered branch locations.